3.2 NATIONAL GEOGRAPHIC EXPLORER PARDIS SABETI



Genetics, Disease, and Native Americans

"The impact that science has on the world around us is something I'm enthralled with." —Pardis Sabeti

Meet National Geographic Explorer **Pardis Sabeti**. She is a research scientist, a musician, a teacher, and a volleyball player. She also doesn't sleep much. When Sabeti is not teaching a class at Harvard University, you might find her analyzing data in her lab, playing bass with her band, collecting virus samples in West Africa, or using mathematics to understand the latest epidemic.

^ Pardis Sabeti, shown here in her Harvard University lab, loves to engage with her students and collaborate with colleagues all over the world to prevent major outbreaks of deadly diseases.

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MAIN IDEA Modern scientific research can help explain how diseases from Europe impacted Native Americans in North America.

REFUGEE TO RESEARCH SCIENTIST

When Pardis Sabeti was two years old, she and her family escaped Iran just before its 1979 revolution and settled in Florida. In school, Sabeti learned that she loved math. That's right: math. She followed her interests into medical school, where she fell in love with research and data analysis. That's right: research and data analysis.

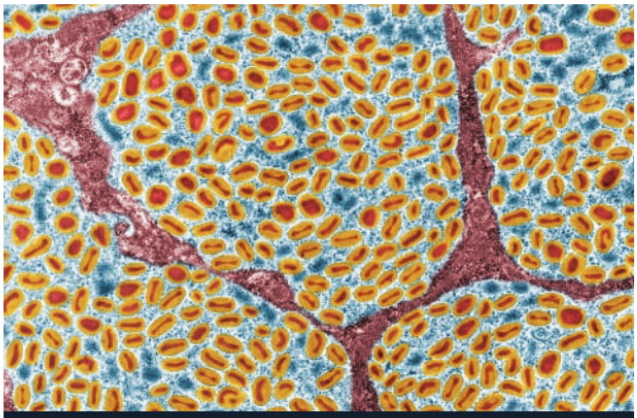
While in graduate school, Sabeti developed a pathbreaking algorithm, or a procedure for solving a problem or analyzing data using a computer. She used this algorithm to analyze a specific gene, or the part of a cell that controls growth, appearance, and traits. She knew she'd made a great discovery. She recalls, "I realized I'd found a trait that had to be the result of natural selection—a trait that likely helped the population I was looking at cope with malaria better than others. It was an amazing feeling because at that moment I knew something about how people evolved that nobody else knew."

Today, Sabeti specializes in the study of infectious diseases, or diseases that spread from person to person. She uses mathematical and computer science tools to analyze the different ways diseases change over time and how they influence changes in human biology. Sabeti works in her lab at Harvard University and in countries impacted by diseases such as malaria and Lassa fever.

UNDERSTANDING DISEASE

Sabeti's research has an urgency, and it's risky. Recently, she led a research team during an outbreak of Ebola in Sierra Leone, a country in West Africa. She and her colleagues determined that the Ebola virus actually spread through human-to-human contact, not from contact with bats or other animals. This knowledge helped health professionals stop, or at least slow down, the epidemic and save many lives.

Her research also reveals how infectious diseases have been some of the most important factors in human history. According to Sabeti, more soldiers have died of infections or from exposure to new climates—and therefore new diseases—than from battle itself. Disease also played an enormous role when Europeans and Native Americans came into contact with each other. Europeans brought diseases to North America, including tuberculosis and smallpox. Europeans likely had inherited immunity, or a genetic protection, against these diseases. But Native Americans had not yet been exposed to them. Smallpox, especially, ravaged Native American populations. This disease also became a weapon of war when European colonizers realized the effect that blankets infected with smallpox could have on native populations.



Doctors dread seeing the smallpox virus, shown here, in patients. Smallpox is highly contagious and could cause a major human catastrophe if uncontrolled.

If you ask Sabeti what she most enjoys about her work, she'd likely respond with more than one answer. She's inspired by colleagues in Africa researching the treatment and prevention of Lassa fever. She thrives on working with her students in classes and in the lab. And she's thrilled to be utilizing her skills in math, research, and data analysis to help develop new treatments for devastating diseases like malaria, a disease that kills more than 1 million people every year.

HISTORICAL THINKING

1. **READING CHECK** What kind of research does Pardis Sabeti do?
2. **SUMMARIZE** In what ways have infectious diseases shaped human history?
3. **MAKE INFERENCES** In what ways might understanding how diseases change over time help doctors prevent future epidemics?

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